

Karthik Sabhanayakam

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EDUCATION

Georgia Institute of Technology *Atlanta, Georgia*
Bachelor of Science in Computer Science

Expected Graduation: December 2027

EXPERIENCE

Marine Robotics Group *Software Engineering Lead*

August 2025 - January 2026

- Programmed autonomous operational capabilities into the "Stinger Tug" marine vehicle project using ROS2 software framework and state-of-the-art computer vision techniques.
- Developed multi-step reasoning capabilities for vehicle navigation, integrating real-time sensor inputs including Lidar imaging for autonomous problem-solving.

San Jose State University *Research Intern*

August 2023 - March 2025

- Investigated environmental patterns across 1,300+ US Superfunds by implementing multimodal machine learning algorithms tailored for complex geospatial and satellite datasets.
- Automated data harvesting using the Google Maps API and geocoded block groups to build a pipeline processing 15,971 images for a high-accuracy ResNet18 classification model.
- Presented peer-reviewed, published findings on geospatial data modeling at international conferences, including IEEE ICMLA (22.5% acceptance rate), ISEC, and SCURR to audiences of over 350+ researchers.

Daewoong Academy *Software Engineering Intern*

May 2024 - December 2024

- Led development of LLM-powered auto-grader, managing data integration pipelines and optimized backend workflows to handle a 2x increase in image-format data formats.
- Architected robust REST API endpoints for online SAT and AP testing platforms, coordinating sprints in Jira.

Audax Labs *Machine Learning Engineering Intern*

October 2023 - February 2024

- Deployed full-stack production application using Docker for real-time water fault detection.
- Engineered autonomous data-parsing pipelines to proactively detect and comprehensively report on over 3,900+ distinct cases of system anomalies and faults for engineering supervisors.

PROJECTS

Mitigating Overfitting in Neural Net Classification of Radio Galaxies through Wavelet Analysis

- Published a machine learning paper introducing a wavelet analysis pipeline that achieved a 10.52% accuracy increase over state-of-the-art convolutional neural networks by eliminating noise artifacts during data ingestion .

Predicting the Instance of Breast Cancer within Patients using a Convolutional Neural Network

- Developed deep learning classification algorithms for breast cancer, presenting findings and technical architecture to an audience of 100+ domain experts.

SKILLS

Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Linear Algebra, Multivariable Calculus, Computer Organization and Programming

Programming Languages: Python · JavaScript · TypeScript · C · Java · SQL

Tools and Frameworks: Docker · MATLAB · Linux · Git · GitHub Actions · NumPy · PyTorch · Pandas · MongoDB · Jira · PCB Design · ROS2 · React · React Native · Next.js · Django · Flask